Status

RK3588 / Orange Pi 5 Max — A/B-virt Emulator — Status

Revision: 3 **Last modified:** 2026-06-11T13:15:00Z **Scope:** The hardware-free emulation ladder that exercises the Helix OTA native Android A/B update flow (`update_engine` + AVB/dm-verity + auto-rollback) for RK3588 / Orange Pi 5 Max targets, on a developer host with NO live board. **Authority:** §11.4.45 (integration Status doc), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing — pending work is stated as PENDING, never implied done), §11.4.107 (liveness, not a single frame), §11.4.112 (structurally-gated tiers SKIP honestly). **HEAD at last update:** `42be557`.

Operator-blocked / Pending items (read first — §11.4.45(9))

Item	State	What is needed	§-ref
A/B slot switch + auto-rollback (the real OTA-apply core)	**PROVEN — PASS (E3, E5)**	DONE. Real U-Boot 2024.01 + QEMU `virt` + HVF: slot A→B select PROVEN (E3, evidence `docs/qa/20260611T094958Z-ab-slot-switch/`, determinism 3/3 identical) + corrupt-slot bootcount→altbootcmd auto-rollback PROVEN (E5, evidence `docs/qa/20260611T095918Z-ab-rollback/`).	§11.4.5
RAUC dm-verity slot integrity	**PENDING_FORENSICS — `ab_rauc_verity.sh` AUTHORED, UNVERIFIED**	NOT yet proven. The RAUC dm-verity driver `tests/emulator/ab_virt/`	§11.4.6

Item	State	What is needed	§-ref
		`ab_rauc_verity.sh` is AUTHORED but UNVERIFIED — gated on a signed `.raucb` bundle + reconciliation of the RAUC slot-class scheme with the proven `boot.cmd` `BOOT_ORDER` env scheme. No captured dm-verity evidence yet.	
Tier-2 Cuttlefish — REAL Android A/B OTA	**SKIP on this host**	No `/dev/kvm` on this Apple-Silicon macOS dev host (confirmed absent). Ready to RUN on the operator's incoming Linux + nested-KVM host. NOT a fake PASS — no release tag while this is SKIP (§11.4.40).	§11.4.3 / §11.4.112
Tier-3 — real RK3588 / Orange Pi 5 Max hardware	**PENDING — no board**	Requires physical hardware on the bench.	§11.4.6

Fidelity ladder (honest)

The emulation tiers trade fidelity for hardware-freedom. Higher tiers exercise more of the real OTA stack but demand scarcer environments.

Tier	What it proves	Fidelity	State
T0 — Containerized control-plane + device emulator (podman)	Control-plane ⇆ device round-trip (register → update-check → telemetry) with NO live hardware	Protocol / control-plane only — no real A/B apply	**SHIPPED**
T1 — A/B-virt base image on QEMU `virt` + HVF	The aarch64 guest boots to a live interactive Linux userspace on the Apple CPU, AND real U-Boot 2024.01	Real kernel + userspace boot; real U-Boot bootcount/ `BOOT_ORDER` slot-select + altbootcmd	**GREEN** — boot + slot-switch (E3) + auto-rollback (E5) all PROVEN; **RAUC dm-verity PENDING** (authored `ab_rauc_verity.sh`,

Tier	What it proves	Fidelity	State
	performs a genuine **A/B slot switch** + **corrupt-slot auto-rollback**	rollback; RAUC dm-verity layer AUTHORED, not yet run	gated on signed bundle)
T2 — Cuttlefish (cvd) on Linux + nested KVM	The **real** Android `update_engine` A/B + AVB/dm-verity + auto-rollback apply flow	Closest hardware-free proxy for the RK3588 OTA apply	**SKIP** on this host (no `/dev/kvm`); ready for Linux host
T3 — Real RK3588 / Orange Pi 5 Max board	The genuine on-device OTA apply + AVB/dm-verity + bootloader rollback	Full fidelity	**PENDING** — no hardware

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Captured-evidence status table (§11.4.5)

Closed verdict vocabulary: `PASS` / `FAIL` / `SKIP` / `PENDING_FORENSICS` / `OPERATOR-BLOCKED`. Every PASS/SKIP cites a real evidence path verified to exist in this repo at revision time.

#	Capability under test	Verdict	Evidence (real, repo-relative)	No
E1	**PWU-AB-0/1 base guest image + real u-`boot.bin` built** (aarch64 Buildroot, kernel 6.1.44 + ext2 rootfs; U-Boot 2024.01)	**PASS**	`tests/emulator/ab_virt/out/images/Image` (MD5 `9f3670cd7dba7bdeeebe2c6d791e929a`), `tests/emulator/ab_virt/out/images/rootfs.ext2` (MD5 `a056760e88eea575977be13e38cfe430`); builder `tests/emulator/ab_virt/build_image.sh`. The U-Boot+RAUC build COMPLETED — a real `u-boot.bin` (`U-Boot 2024.01 (Jun 11 2026 - 08:21:42 +0000)`, per the E3/E5 verdict files) now drives the A/B boot.	The "bu not RE boo and swi rol flo it.
E2	**PWU-AB-1 FOUNDATION — boots to LIVE interactive userspace on QEMU `virt` + HVF**	**PASS**	`docs/qa/20260611T061626Z-ab-virt-boot/console.log` (196 lines) — kernel boots on Apple CPU (`physical CPU 0x0000000000 [0x610f0000]`, MIDR `0x610f`), `buildroot login: root` (line 174), post-login sentinel `HELIX_USERSPACE_LIVE_OK` (line 182, emitted only from an interactive shell after login), clean `poweroff`.	§11 live boo pos sen sin Dri emu `ab_` `boo`

#	Capability under test	Verdict	Evidence (real, repo-relative)	No
E3	**A/B slot switch (slot A→B select)**	**PASS**	`docs/qa/20260611T094958Z-ab-slot-switch/` — `verdict.txt` (Run A rc=0 → `HELIX_SLOTID=A` `/dev/vda2`; Run B rc=0 → `HELIX_SLOTID=B` `/dev/vda3`, the slot genuinely SWITCHED); per-run console transcripts `consoleA.log` / `consoleB.log`; `determinism_n3.txt` (**3/3 identical PASS** iterations, §11.4.50). Driver `tests/emulator/ab_virt/ab_slot_switch.sh` + `uboot_ab/boot.cmd`.	PR aga Bo QE HV liv run pos sen int (no fra fir dev con ker hel the NO oth (th rea op)
E4	**RAUC dm-verity integrity on the booted slot**	**PENDING_FORENSICS**	RAUC dm-verity driver `tests/emulator/ab_virt/ab_rauc_verity.sh` AUTHORED — NOT yet run	UN Ga sig bur slo boo BOO env rec Ho per no
E5	**Auto-rollback on corrupt slot (bootcount > bootlimit → altbootcmd swap → known-good slot)**	**PASS**	`docs/qa/20260611T095918Z-ab-rollback/` — `verdict.txt` (Run ROLLBACK rc=0: head=bad slot B, `bootcount=2 > bootlimit=1` → altbootcmd swap → booted slot A `/dev/vda2`; Run CONTROL rc=0: head slot B, no rollback → B `/dev/vda3`); console transcripts `consoleROLLBACK.log` (shows `A/B: bootcount=2 > bootlimit=1 -> rolling back (altbootcmd swap)` → `booting slot` A, guest `HELIX_SLOTID=A`) / `consoleCONTROL.log`. Driver `tests/emulator/ab_virt/ab_rollback.sh` + `uboot_ab/boot.cmd`.	PR aga Bo QE HV CO is t pro rol fire slo unc

#	Capability under test	Verdict	Evidence (real, repo-relative)	No
E6	**T0 containerized control-plane ⇆ device round-trip**	**PASS**	Driver `tests/emulator/tier1_container_e2e.sh` (boots `ota-server` + `ota-device-emu` in a podman pod; asserts register → update-check → telemetry from captured container logs under `docs/qa/<run-id>/`)	Pro pla onl NO rea
E7	**T2 Cuttlefish — REAL Android A/B OTA apply (incl. PWU-CF-2 corrupt-slot auto-rollback)**	**SKIP — UNVERIFIED pending Linux host**	Driver `tests/emulator/tier2_cuttlefish_ab.sh` (PWU-CF-2 headline corrupt-slot → reboot → auto-rollback section authored, mirrors ab_virt PWU-AB-3); host gate: `/dev/kvm` absent on this Apple-Silicon macOS host (verified)	§11 §11 top (ex fak scr ma OT rol inv `UNC` per Lin run the inc hos
E8	**T3 real RK3588 / Orange Pi 5 Max hardware**	**PENDING_FORENSICS**	—	No ben
E9	**HelixQA bank — `helix_ota.yaml` static gate (incl. `HOTA-AB-VIRT-BOOT`)**	**PASS**	`bash tools/helixqa/run_bank.sh --dry-run --bank tools/helixqa/banks/helix_ota.yaml` → **12 passed** / **0 failed** / **0 skipped**; `HOTA-AB-VIRT-BOOT` declares evidence `docs/qa/20260611T061626Z-ab-virt-boot/console.log` (the same E2 boot transcript, verified present)	Sta arti run any wh evi mis NC exe app tha evi is v bar

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What is genuinely proven today (§11.4.6)

- **Proven (captured evidence):** the A/B-virt base guest image **builds** and **boots to a live, interactive Linux userspace** on QEMU `virt` + HVF on this Apple-Silicon host (E1, E2) — the foundation tier — and a real `u-boot.bin` (U-Boot 2024.01) now drives the boot.

- **Proven (captured evidence) — the A/B-apply core:** the real **A/B slot switch** (E3) and the **corrupt-slot auto-rollback** (E5) are PROVEN against real U-Boot 2024.01 + QEMU `virt` + HVF. E3 (`docs/qa/20260611T094958Z-ab-slot-switch/`) shows slot A and slot B both selected and booted, the guest userspace reporting the switched `slot_id`, the root device confirmed by `findmnt`, and the negative "did NOT boot the other slot" check — replayed **3/3 identical** (§11.4.50). E5 (`docs/qa/20260611T095918Z-ab-rollback/`) shows U-Boot `bootcount=2 > bootlimit=1` triggering the `altbootcmd` swap that falls back from the bad slot B to the known-good slot A, with a CONTROL run proving the rollback only fires on a bad slot.
- **AUTHORED but NOT run (honestly pending):** RAUC dm-verity slot integrity (E4). The driver `tests/emulator/ab_virt/ab_rauc_verity.sh` is written but UNVERIFIED — gated on a signed `.raucb` bundle + reconciling the RAUC slot-class scheme with the proven `boot.cmd` `BOOT_ORDER` env scheme. No dm-verity evidence yet, so it is not marked proven.
- **Bank static gate (proven):** the HelixQA `helix_ota.yaml` bank passes its `--dry-run` evidence-artifact audit 12/0/0 (E9), with `HOTA-AB-VIRT-BOOT` wired to the verified E2 boot transcript.
- **Honestly skipped (topology-gated):** the Tier-2 Cuttlefish real-Android-A/B path including the PWU-CF-2 corrupt-slot auto-rollback section (E7) SKIPs on this host for lack of `/dev/kvm` and is UNVERIFIED-pending a Linux + nested-KVM host. No fabricated continuity, no fake PASS.
- **No release tag** — §11.4.40 requires the full ladder GREEN; T2 Cuttlefish and T3 hardware remain SKIP/PENDING, so a release tag would be a bluff.

Provenance

Commit	Subject
`dd43738`	research + dev-host A/B-virt build infra (base image build in progress)
`d5374d0`	PWU-AB-1 foundation GREEN — base image boots to LIVE userspace on QEMU+HVF
`4278aa9`	add U-Boot qemu_arm64 + RAUC + GPT tooling to the A/B-virt build (PWU-AB-1 full, in progress)
`0f120a6`	parallel streams — HelixQA bank wiring (`HOTA-AB-VIRT-BOOT`) + these §11.4.45 Status docs + Cuttlefish corrupt-slot rollback (PWU-CF-2)
`6581ab4`	A/B disk assembler (`assemble_ab_disk.sh`) + U-Boot bootcount slot-select (`uboot_ab/boot.cmd`) — PWU-AB-1, AUTHORED
`0d27438`	U-Boot host tools need `libssl-dev` (+bison/flex) — build7 failed on missing `openssl/evp.h`
`18ed84a`	

Commit	Subject
	PWU-AB-1: REAL U-Boot A/B slot switch PROVEN on macOS QEMU+HVF (E3, evidence `docs/qa/20260611T094958Z-ab-slot-switch/`)
`afcad1d`	§11.4.50 determinism soak + PWU-AB-2 (`ab_rauc_verity.sh`) / PWU-AB-3 authored + docs/bank
`42be557`	**PWU-AB-3: REAL corrupt-slot AUTO-ROLLBACK proven on macOS QEMU+HVF** (E5, evidence `docs/qa/20260611T095918Z-ab-rollback/`) — current HEAD